

SEULCHAN LEE

E seul94@skku.edu; leesc1207@gmail.com

T +82-31-290-7543; +82-10-5245-4651

[Google Scholar](#); [Personal Website](#)

Postdoctoral Researcher

Construction and Environmental Research Center

Earth and Remote Sensing Lab.

Sungkyunkwan University (SKKU)

2066 Seobu-ro, Suwon 16419, Republic of Korea

EDUCATION

2019 – 2025 **Ph.D. in Water Resources**

Sungkyunkwan University, Suwon, Republic of Korea (advisor: [Dr. Minha Choi](#))

Dissertation: *Integrating Intermediate-Scale Observations and Land Surface Modeling for Synthetic Aperture Radar Soil Moisture Retrieval* [\[Link\]](#)

2019 **B.S. in Civil and Environmental Engineering**

Sungkyunkwan University, Suwon, Republic of Korea

ACADEMIC & PROFESSIONAL EXPERIENCE

2025 – now **Postdoctoral Researcher** (advisor: [Dr. Minha Choi](#))

Construction and Environmental Research Center, Sungkyunkwan University, Suwon, Republic of Korea

2026 **Lecturer**

Dept. of Water Resources, Sungkyunkwan University, Suwon, Republic of Korea:

- (GWR5048) *Statistical Hydrology*, Spring Sems. 2026

2019 – 2025 **Graduate Research Assistant**

Dept. of Water Resources, Sungkyunkwan University, Suwon, Republic of Korea

2019 – 2024 **Teaching Assistant**

School of Civil, Architectural Eng. & Landscape Architecture, Dept. of Water Resources, Dept. of Civil and Environmental Engineering, Sungkyunkwan University:

- (CAL3086) *Remote Sensing of Global Environment*, Fall Sems. 2025
- (CAL3086) *Remote Sensing of Global Environment*, Fall Sems. 2024
- (GWR5047) *Water Resources & Remote Sensing*, Spring Sems. 2020
- (GWR5048) *Statistical Hydrology*, Fall Sems. 2019
- (CAL2015) *Introduction to Environmental Engineering*, Fall Sems. 2019
- (CAL3037) *Hydrology*, Spring Sems. 2019
- (CAL2014) *Fundamentals of Environmental Engineering and Science*, Spring Sems. 2019

RESEARCH INTERESTS

- Water, Ecosystem and Environmental Science; Land surface-atmosphere interactions; Remote sensing
- Soil moisture dynamics; Evapotranspiration; Rainfall-Runoff simulation; Land surface modeling
- Flood and Drought analysis; Data Assimilation

HONORS & AWARDS

2024 Dean's List, Sungkyunkwan University (1st semester)

2019 – 2021 Outstanding Graduate Student Scholarship, Sungkyunkwan University

2019 – 2020 Teaching and Research Assistant Scholarship, Sungkyunkwan University

2018 Dean's List, Sungkyunkwan University (1st, 2nd semester)

2017 Dean's List, Sungkyunkwan University (2nd semester)

2013 – 2018 National Science & Technology Scholarship, Korea Student Aid Foundation (KOSAF)

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES (SCI)

- 2026 M. Kang, **S. Lee (co-first)**, D. Kim, and M. Choi*. 2026. Hydroclimatic Controls on Flash Drought Development in Northeast Asia: Roles of Atmospheric Demand and Water Availability. *Hydrological Processes*. 40(7), e70618. <https://doi.org/10.1002/hyp.70618>
- 2026 D. Kim, **S. Lee**, and M. Choi*. 2026. Global CMIP6-based datasets of vegetation-responsive and feedback-deflated potential evapotranspiration. *Scientific Data*. <https://doi.org/10.1038/s41597-026-07553-0>
- 2026 **S. Lee**, S. Cho, and M. Choi*. 2026. A novel approach for soil moisture retrieval from Sentinel-1 SAR via temporal stability-based backscatter analysis. *Remote Sensing of Environment*. 337, 115319. <https://doi.org/10.1016/j.rse.2026.115319>
- 2025 D. Kim, **S. Lee**, S. Cho, D. Kim, and M. Choi*. 2025. Evaluating rainfall estimates derived from soil moisture using soil hydraulic properties over the Korean Peninsula. *Journal of Hydrology*. 663B, 134267. <https://doi.org/10.1016/j.jhydrol.2025.134267>
- 2025 **S. Lee**, J. Park, J. Jeong, and M. Choi*. 2025. Soil moisture and evapotranspiration responses to precipitation uncertainty using Noah-MP land surface model. *Stochastic Environmental Research and Risk Assessment*. 2025, 1-20. <https://doi.org/10.1007/s00477-025-02991-5>
- 2024 W. Kim, **S. Lee**, and M. Choi*. 2024. A comparative study of remotely sensed reservoir monitoring across multiple land cover types. *Science of the Total Environment*. 948, 174678. <https://doi.org/10.1016/j.scitotenv.2024.174678>
- 2024 J. Jeong, **S. Lee**, S. Cho, K. Kim, and M. Choi*. 2024. Multi-scale assessment of a cosmic-ray neutron probe observation of soil moisture for surface layer applications in a mountainous forest environment. *Agricultural and Forest Meteorology*. 356, 110155. <https://doi.org/10.1016/j.agrformet.2024.110155>
- 2023 **S. Lee**, S. Oh, R.L. Ray, Y. Lee, and M. Choi*. 2023. Three-dimensional hydrological thresholds to predict shallow landslides. *Terrestrial, Atmospheric and Oceanic Sciences*. 34, 20(2023). <https://doi.org/10.1007/s44195-023-00052-4>
- 2021 J. Jeong, **S. Lee**, and M. Choi*. 2021. Correction efficiency and error characteristics for cosmic-ray soil moisture on mountainous terrain. *Journal of Hydrology*. 601, 126657. <https://doi.org/10.1016/j.jhydrol.2021.126657>

ONGOING WORKS & UNDER REVIEW

- 2026 S. Cho, **S. Lee (co-first)**, D. Kim, and M. Choi*. 2026. Soil moisture retrieval from Sentinel-1 SAR using a water cloud model-constrained physics-informed neural network. *GIScience & Remote Sensing*. **Under review** (Jun. 2026).
- 2026 W. Kim, S. Cho, **S. Lee**, H. Jeon, J.M. Jacobs, and M. Choi*. 2026. Extraction of Water Bodies from SAR Imagery and Swin Transformer Model in Diverse Hydrological Conditions. *In Preparation*.
- 2026 **S. Lee**, D. Kim, K. Kim, and M. Choi*. 2026. CRNP as the Regional Truth: Assessing the Representativeness of In-Situ Sensors for CRNP Calibration in Heterogeneous Terrains. *In Preparation*.

PEER-REVIEWED KOREAN JOURNAL ARTICLES (KCI)

- 2024 S. Lee, W. Kim, **S. Lee**, H. Jeong, J. Park, and M. Choi*. 2024. Comparative study of flood detection methodologies using Sentinel-1 satellite imagery. *Journal of Korea Water Resources Association*. 57(3), 181-193. <https://doi.org/10.3741/JKWRA.2024.57.3.181>
- 2023 M. Kang, J. Jeong, **S. Lee**, and M. Choi*. 2023. A Study on Spatial Onset Characteristics of Flash Drought based on GLDAS Evaporative Stress in the Korean Peninsula. *Journal of Korea Water Resources Association*. 56(10), 631-639. <https://doi.org/10.3741/JKWRA.2023.56.10.631>
- 2023 H. Jeon, **S. Lee**, Y. Lee, J. Kim, and M. Choi*. 2023. Evaluation of Bias and Uncertainty in Snow Depth Reanalysis data over South Korea. *Journal of Korea Water Resources Association*. 56(9), 543-551. <https://doi.org/10.3741/JKWRA.2023.56.9.543>
- 2023 **S. Lee**, W. Kim, S. Cho, H. Jeon, and M. Choi*. 2023. Water Resources Monitoring Technique using Multi-source Satellite Image Data Fusion. *Journal of Korea Water Resources Association*. 56(8), 497-508.

- <https://doi.org/10.3741/jkwra.2023.56.8.497>
- 2023 J. Jeong, S. Cho, **S. Lee**, K. Kim, Y. Lee, C.D. Lee, S. Lee, and M. Choi*. 2023. A Study on Domestic Applicability for the Korean Cosmic-Ray Soil Moisture Observing System. *Korean Journal of Remote Sensing*. 39(2), 233-246. <https://doi.org/10.7780/kjrs.2023.39.2.9>
- 2022 **S. Lee**, J. Jeong, S. Oh, H. Jeong, and M. Choi*. 2022. Multi-resolution SAR Image-based Agricultural Reservoir Monitoring. *Korean Journal of Remote Sensing*. 38(5), 497-510. <https://doi.org/10.7780/kjrs.2022.38.5.1.5>
- 2022 J. Jeong, S. Cho, H. Jeon, **S. Lee**, and M. Choi*. 2022. Assessment of Stand-alone Utilization of Sentinel-1 SAR for High Resolution Soil Moisture Retrieval Based on Machine Learning. *Korean Journal of Remote Sensing*. 38(5), 571-585. <https://doi.org/10.7780/kjrs.2022.38.5.1.11>
- 2022 S. Choi, Y. Youn, J. Kang, G. Park, G. Kim, **S. Lee**, M. Choi, H. Jeong, and Y. Lee*. 2022. An Artificial Intelligence Approach to Waterbody Detection of the Agricultural Reservoirs in South Korea Using Sentinel-1 SAR Images. *Korean Journal of Remote Sensing*. 38(5), 925-938. <https://doi.org/10.7780/kjrs.2022.38.5.3.10>
- 2022 K. Kim, **S. Lee**, Y. Lee, M. Yeon, G. Lee, and M. Choi*. 2022. Surface soil moisture memory using stored precipitation fraction in the Korean peninsula. *Journal of Korea Water Resources Association*. 55(2), 111-120. <https://doi.org/10.3741/jkwra.2022.55.2.111>
- 2021 J. Jeong, S. Oh, **S. Lee**, J. Kim, and M. Choi*. 2021. Sentinel-1 SAR image-based waterbody detection technique for estimating the water storage in agricultural reservoirs. *Journal of Korea Water Resources Association*. 54(7), 535-544. <https://doi.org/10.3741/jkwra.2021.54.7.535>
- 2021 S. Cho, J. Jeong, **S. Lee**, and M. Choi*. 2021. Estimation of soil moisture based on Sentinel-1 SAR data: Assessment of soil moisture estimation in different vegetation condition. *Journal of Korea Water Resources Association*. 54(2), 81-91. <https://doi.org/10.3741/jkwra.2021.54.2.81>
- 2020 S. Oh, J. Jeong, **S. Lee**, and M. Choi*. 2020. Downscaling GPM Precipitation Using Finer-scale MODIS Based Optical Image in Korean Peninsula. *Korean Journal of Remote Sensing*. 36(5), 749-762. <https://doi.org/10.7780/kjrs.2020.36.5.1.8>
- 2020 S. Cho, J. Jeong, **S. Lee**, and M. Choi*. 2020. Estimation of soil moisture based on sentinel-1 SAR data: focusing on cropland and grassland area. *Journal of Korea Water Resources Association*. 53(11), 973-983. <https://doi.org/10.3741/jkwra.2020.53.11.973>
- 2020 H. Jeon, J. Baik, **S. Lee**, and M. Choi*. 2020. A point-scale gap filling of the flux-tower data using the artificial neural network. *Journal of Korea Water Resources Association*. 53(11), 929-938. <https://doi.org/10.3741/jkwra.2020.53.11.929>
- 2019 **S. Lee**, J. Baik, M. Choi, and Y. Cho*. 2019. Evaluation of the behavior and quality in soil moisture data: A case study of Yongdam study watershed. *Journal of Korea Water Resources Association*. 52(12), 951-962. <https://doi.org/10.3741/jkwra.2019.52.12.951>
- 2019 **S. Lee**, J. Jeong, J. Park, H. Jeon, and M. Choi*. 2019. Estimating Fine Particulate Matter Concentration using GLDAS Hydrometeorological Data. *Korean Journal of Remote Sensing*. 35(6-1), 919-932. <https://doi.org/10.7780/kjrs.2019.35.6.1.4>
- 2019 J. Baik, S. Cho, **S. Lee**, and M. Choi*. 2019. Analysis on Adequacy of the Satellite Soil Moisture Data (AMSR2, ASCAT, and ESACCI) in Korean Peninsula: With Classification of Freezing and Melting Periods. *Korean Journal of Remote Sensing*. 35(5-1), 625-636. <https://doi.org/10.7780/kjrs.2019.35.5.1.1>

ARTICLES IN PROFESSIONAL MAGAZINES

- 2026 **S. Lee** and M. Choi. 2026. Overcoming the spatial representativeness limitations in remote sensing hydrology: validation of high-resolution SAR soil moisture retrievals using cosmic-ray neutron sensing (CRNS). *Water for Future* (Korea Water Resources Association).
- 2022 **S. Lee**, M. Kang, and M. Choi. 2022. Reservoir water body monitoring using multi-satellite images. *Water for Future* (Korea Water Resources Association). 55(12), 6-10. [\[Link\]](#)
- 2022 S. Oh, **S. Lee**, W. Kim, and M. Choi. 2022. Water body detection technique using Sentinel-1 SAR images. *Water for Future* (Korea Water Resources Association). 55(6), 47-51. [\[Link\]](#)
- 2021 J. Jeong, S. Cho, **S. Lee**, S. Oh, H. Jeon, W. Kim, and M. Choi. 2021. Artificial Intelligence, a future partner for hydrological analyses. *Water for Future* (Korea Water Resources Association). 54(12), 8-14. [\[Link\]](#)

CONFERENCE PROCEEDINGS, POSTERS, AND PRESENTATIONS

INTERNATIONAL CONFERENCES

- 2026 D. Kim, S. Lee, J. Jeong, S. Cho, and M. Choi. 2026. Exploring Three-dimensional Soil Moisture Variability using Multi-Source Observations. *EGU General Assembly 2026*, 2026.05.03-08, Vienna, Austria
- 2026 J. Jeong, D. Kim, S. Lee, W. Kim, and M. Choi. 2026. Integration of SMAP L-band VOD and Multi-source Satellite Data for Improved Sentinel-1 Soil Moisture Retrieval in Complex Terrains. *EGU General Assembly 2026*, 2026.05.03-08, Vienna, Austria
- 2025 D. Kim, S. Cho, S. Lee, and M. Choi. 2025. Spatiotemporal Variability of Soil Moisture: Multi-Sensor Observations at the Eoseom Korea Hydrometeorological Observatory. *AGU 2025 Fall Meeting*, 2025.12.15-19, Online
- 2025 J. Jeong, S. Lee, D. Kim, and M. Choi. 2025. High-Resolution Root Zone Soil Moisture Modeling for Small Watersheds using the Land Information System and Multi-Product Comparison. *2025 KAGIS Autumn Conference & IAG'International Symposium*, 2025.11.06-08, Jeju, Republic of Korea
- 2025 S. Lee, J. Jeong, and M. Choi. 2025. Improving Soil Moisture Retrieval Accuracy with Temporal Stability-based Masking of SAR backscatter. *SPIE Sensors + Imaging 2025*, 2025.09.15-18, Madrid, Spain
- 2025 H. Jeon, S. Lee, and M. Choi. 2025. High-Resolution Evapotranspiration Modeling from Satellite and Drone Observations via METRIC. *SPIE Sensors + Imaging 2025*, 2025.09.15-18, Madrid, Spain
- 2025 J. Jeong, S. Lee, D. Kim, and M. Choi. 2025. Prediction and Evaluation of Hydrological Factors and Drought Indices under Climate Change using CMIP6 in Monsoon Regions. *EGU General Assembly 2025*, 2025.04.27-05.02, Vienna, Austria
- 2024 M. Kang, S. Lee, and M. Choi. 2024. Role of Terrestrial Evapotranspiration on the Occurrence of Flash Droughts in Northeast Asia. *IGARSS 2024*, 2024.07.07-12, Athens, Greece
- 2024 S. Cho, S. Lee, H. Jeon, and M. Choi. 2024. SAR Soil Moisture Retrieval using Physically based AI Model. *IGARSS 2024*, 2024.07.07-12, Athens, Greece
- 2024 S. Lee, J. Park, and M. Choi. 2024. Precipitation Forcing-induced Uncertainties in Land Surface Model (LSM) Simulations. *AOGS 2024 21st Annual Meeting*, 2024.06.23-28, Pyeongchang, Republic of Korea
- 2024 H. Jeon, S. Lee, and M. Choi. 2024. Evaluation of LDAPS Snow information by MERRA-2 and ASOS over the South Korea. *EGU General Assembly 2024*, 2024.04.14-19, Vienna, Austria
- 2023 S. Lee, J. Jeong, N.S.M. Harmay, and M. Choi. 2023. Impacts of Precipitation Forcing on Hydrological Simulation over Monsoon Asia. *EGU General Assembly 2023*, 2023.04.23-28, Vienna, Austria
- 2023 W. Kim, S. Lee, and M. Choi. 2023. Flooded area monitoring using SAR image-based water body detection technique. *EGU General Assembly 2023*, 2023.04.23-28, Vienna, Austria
- 2023 J. Jeong, S. Cho, S. Lee, K. Kim, C.D. Lee, S. Lee, and M. Choi. 2023. Evaluation of Cosmic-Ray Neutron Probe Based Soil Moisture Observatory for Korean cOsmic-ray Soil Moisture Observing System (KOSMOS). *The 28th International Symposium on Remote Sensing*, 2023.04.19-21, Jeju, Republic of Korea
- 2023 S. Cho, J. Jeong, S. Lee, and M. Choi. 2023. Soil Moisture Estimation and Mapping using Sentinel-1 on vegetated area in Korean Peninsula using Physical methods. *The 28th International Symposium on Remote Sensing*, 2023.04.19-21, Jeju, Republic of Korea
- 2023 W. Kim, S. Lee, and M. Choi. 2023. Agricultural reservoir monitoring using multi-dimensional satellite images. *The 28th International Symposium on Remote Sensing*, 2023.04.19-21, Jeju, Republic of Korea
- 2022 S. Lee, J. Park, J. Jeong, and M. Choi. 2022. Influence of Precipitation Forcing in Hydrological Simulation over Monsoon Asia. *AGU 2022 Fall Meeting*, 2022.12.12-16, Online
- 2022 S. Cho, S. Lee, S. Oh, and M. Choi. 2022. High resolution soil moisture estimation considering complex vegetated areas with modified Water Cloud Model. *AGU 2022 Fall Meeting*, 2022.12.12-16, Online
- 2022 S. Oh, S. Lee, W. Kim, and M. Choi. 2022. The use of Synthetic Aperture Radar for surface waterbody detection: a comparative study of thresholding methods. *AGU 2022 Fall Meeting*, 2022.12.12-16, Online
- 2021 M. Choi, S. Lee, J. Jeong, and S. Cho. 2021. Conditional Gap-filling of in-situ Soil Moisture for Continuous Water Monitoring. *IWRA World Water Congress*, 2021.11.29-12.03, Daegu, Republic of Korea
- 2020 M. Choi, H.H. Nguyen, J. Jeong, S. Lee, S. Oh, and S. Cho. 2020. Assessment of the soil moisture-

evaporative fraction coupling strength using the Cosmic-Ray Neutron Sensing network. *AGU 2020 Fall Meeting*, 2020.12.07-11, Online

- 2020 **S. Lee**, H. Jeon, J. Park, and M. Choi. 2020. Performance of a Physically Based Gap-Filling Technique of in-situ Soil Moisture, in Comparison with Machine Learning. *EGU General Assembly 2020*, 2020.05.04-08, Online
- 2019 S. Cho, M. Zohaib, **S. Lee**, and M. Choi. 2019. Quantification of Irrigation water consumption using remote sensing dataset in Korea Peninsula. *AGU 2019 Fall Meeting*, 2019.12.09-13, San Francisco, U.S.
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. Statistical Assessment of the Influence of Hydrometeorological factors on Fine Dust. *Asian Conference on Remote Sensing 2019*, 2019.10.14-18, Daejeon, Republic of Korea
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. Hydrometeorological drivers of particulate matter using Bayesian model averaging. *IGARSS 2019*, 2019.07.28-08.02, Yokohama, Japan
- 2018 J. Baik, J. Park, **S. Lee**, U. Kim, and M. Choi. 2018. Assessment of merging technique using Triple Collocation (TC) from satellite and reanalysis dataset over Different Land Covers in East Asia: GLDAS, MOD16, GLEAM, and MERRA. *AGU 2018 Fall Meeting*, 2018.12.10-15, Washington D.C., U.S.

KOREAN CONFERENCES

- 2026 J. Jeong, D. Kim, **S. Lee**, and M. Choi. 2026. Application of SMAP L-band VOD as a Vegetation Indicator in the Water Cloud Model for Sentinel-1 Soil Moisture Estimation over Complex Terrain. *The Korea Water Resources Association Conference*, 2026.05.20-22, Busan
- 2025 **S. Lee** and M. Choi. 2025. Temporal Stability-informed Backscatter Masking Strategy for Sentinel-1 SAR Soil Moisture Retrieval. *The Korean Society of Remote Sensing Conference*, 2025.10.22-24, Yeosu
- 2025 J. Jeong, **S. Lee**, D. Kim, and M. Choi. 2025. Calibration of CRNP Sensor Using Noah-MP LSM-simulated Soil Moisture Data. *The Korean Society of Remote Sensing Conference*, 2025.10.22-24, Yeosu
- 2025 **S. Lee**, S. Cho, and M. Choi. 2025. Enhancing high-resolution soil moisture retrieval from Sentinel-1 SAR: A temporal stability-based approach. *The Korea Water Resources Association Conference*, 2025.05.21-23, Yeosu
- 2025 S. Cho, **S. Lee**, and M. Choi. 2025. The SAR based regional scale soil moisture retrieval and evaluation using land surface information with Physics Informed Neural Network model. *The Korea Water Resources Association Conference*, 2025.05.21-23, Yeosu
- 2025 J. Jeong, **S. Lee**, and M. Choi. 2025. Projection of future drought indices through validation and bias correction of hydrological variables in East Asia using CMIP6. *The Korea Water Resources Association Conference*, 2025.05.21-23, Yeosu
- 2025 M. Choi, **S. Lee**, S. Cho, J. Yi, and S.J. Oh. 2025. A Study on SAR Soil Moisture Retrieval using Physics-based AI model. *Korean Society of Hazard Mitigation Conference*, 2025.02.19-21, Jeju
- 2024 W. Kim, **S. Lee**, and M. Choi. 2024. Multi-satellite remote sensing-based agricultural reservoir monitoring. *The Korean Society of Remote Sensing Conference*, 2024.10.16-18, Seoul
- 2024 M. Choi, W. Kim, and **S. Lee**. 2024. Remotely sensed agricultural reservoir monitoring in Korean peninsula. *The Korean Society of Civil Engineers Conference*, 2024.10.16-18, Jeju
- 2023 **S. Lee**, W. Kim, S. Lee, and M. Choi. 2023. Reservoir water body detection using multi-satellite remote sensing data in Korea. *The Korean Society of Remote Sensing Conference*, 2023.11.15-17, Gyeongju
- 2023 D. Kim, J. Jeong, H. Jeon, **S. Lee**, and M. Choi. 2023. Improving soil moisture accuracy in ungauged areas using Multi-Satellite data. *The Korea Water Resources Association Conference*, 2023.05.25-26, Gosung
- 2022 M. Kang, S. Cho, **S. Lee**, and M. Choi. 2022. Flash drought and its impact on eco-hydrology using MODIS over the Korean peninsula. *The Korean Association of Geographic Information Studies Conference*, 2022.11.03-05, Jeju
- 2022 M. Choi, **S. Lee**, W. Kim, and S. Oh. 2022. Comparative analysis of thresholding techniques for SAR based waterbody detection. *The Korean Society of Remote Sensing Conference*, 2022.11.07-09, Busan
- 2021 N.S.M. Harmay, **S. Lee**, and M. Choi. 2021. Characteristics of the Land Surface Variables at Multiple Land Cover Types under Summer Heatwaves Influences. *The Korean Society of Remote Sensing Conference*, 2021.10.20-22, Online
- 2021 W. Kim, **S. Lee**, J. Park, S. Fred, and M. Choi. 2021. Hydrological drought analysis using SAR image. *The*

Korean Society of Remote Sensing Conference, 2021.10.20-22, Online

- 2021 M. Yeon, L.N. Van, **S. Lee**, and G. Lee. 2021. Analysis of soil erosion in agricultural lands using physics-based erosion model. *The Korea Water Resources Association Conference*, 2021.06.03-04, Gwangju
- 2021 K. Kim, **S. Lee**, M. Choi, S. Jung, and M. Yeon. 2021. Evaluation of multiple-satellite precipitation data by rainfall intensity. *The Korea Water Resources Association Conference*, 2021.06.03-04, Gwangju
- 2020 S. Cho, J. Jeong, **S. Lee**, and M. Choi. 2020. Estimation and mapping of high resolution soil moisture using Sentinel-1 SAR images. *The Korean Society of Remote Sensing Conference*, 2020.11.04-06, Gangchon
- 2019 J. Jeong, **S. Lee**, S. Cho, and M. Choi. 2019. An Analysis on the Relation between Particulate Matter and Hydrometeorological Factors Using Bayesian Model Averaging. *The Korean Society of Civil Engineers Conference*, 2019.10.16-18, Pyeongchang
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. Hydrometeorological Drivers of Particulate Matter Using Satellite and Reanalysis Data. *The Korea Water Resources Association Conference*, 2019.05.30-31, Yeosu
- 2019 **S. Lee**, J. Jeong, and M. Choi. 2019. A Study on the Hydrometeorological Drivers of Particulate Matter. *The Korean Meteorological Society Conference*, 2019.04.22-23, Daegu
- 2018 **S. Lee**, J. Jeong, and M. Choi. 2018. A Study on the Relationship between Precipitation using GPM and Fine Dust Concentration in Spring Season. *The Korean Society of Remote Sensing Conference*, 2018.10.24-26, Muju

RESEARCH PROJECT EXPERIENCES

- 2025 – now Net-zero strategy for sustainable water conservation and nature-based solutions, National Research Foundation of Korea (NRF), Postdoctoral Researcher
- 2024 – 2025 Net-zero strategy for sustainable water conservation and nature-based solutions, National Research Foundation of Korea (NRF), Research Assistant
- 2024 – 2025 Improving storage capacity of underground infrastructure in response to urban flooding and developing multi-functional utilization technology, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
- 2024 – 2025 Cycle adaptive coastal erosion management technology development, Korea Institute of Marine Science & Technology Promotion (KIMST), Research Assistant
 - 2024 Research on the time series analysis techniques for national water surface area using medium-resolution SAR satellite imagery, National Disaster Management Research Institute (NDMI), Research Assistant
- 2024 – 2025 Development of AI-based water resources big data quality management technology, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
- 2023 – 2025 Development of damage analysis technology of water shortage and socio-economic value evaluation technology of water supply, and its application strategy, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
 - 2023 Development of technology for nationwide reservoir surface area analysis and visualization using satellite data, National Disaster Management Research Institute (NDMI), Research Assistant
- 2022 – 2025 Development of comprehensive land management technology using satellite image information bigdata, Korea Agency for Infrastructure Technology Advancement (KAIA), Research Assistant
- 2022 – 2025 Highly precise 3-Dimensional spatial soil moisture tracing algorithm using artificial intelligence for short-term flood and drought prediction, National Research Foundation of Korea (NRF), Research Assistant
- 2022 – 2024 Development of big data-based disaster impact assessment technology for development projects, Ministry of the Interior and Safety (MOIS), Research Assistant
 - 2022 Development of improved waterbody extraction technology using satellite radar data, National Disaster Management Research Institute (NDMI), Research Assistant
- 2021 – 2025 Development of intelligent disaster observation and surveillance technology using satellite imagery, Ministry of the Interior and Safety (MOIS), Research Assistant
- 2021 – 2022 Development of a physics-based modeling system for analysis and prediction of high-risk surface soil failure area, Korea Environmental Industry and Technology Institute (KEITI), Research Assistant
 - 2021 Research on water management strategies based on national satellite imagery (Supporting the

implementation of the Korea-Uzbekistan governmental space cooperation MoU), Korea Aerospace Research Institute (KARI), Research Assistant

- 2021 Development of the multi-satellite information supply and demand system for disaster safety management, National Disaster Management Research Institute (NDMI), Research Assistant
- 2019 – 2022 Blending of multinational/multispectral satellite-based soil moisture products for natural disaster forecast and analysis over Korean peninsula, National Research Foundation of Korea (NRF), Research Assistant
- 2018 – 2021 Development of the merged drought monitoring technology for natural disaster mitigation of satellite-based big data, National Research Foundation of Korea (NRF), Research Assistant
- 2018 – 2019 Development of integrated soil moisture measurement system for national water management, Korea Agency for Infrastructure Technology Advancement (KAIA), Research Assistant
- 2018 – 2019 Improving prediction of Asian dust events using COMS/Metop-A dataset, National Research Foundation of Korea (NRF), Research Assistant
- 2018 – 2019 Blending of multinational/multispectral satellite-based soil moisture products for natural disaster forecast and analysis over Korean peninsula, National Research Foundation of Korea (NRF), Research Assistant

SOFTWARE & SYSTEM DEVELOPMENT

- 2021 Hydrometeorological data inventory system for satellite data hub, Supported by LINC+ Project, Sungkyunkwan University
- 2020 Development of satellite data sharing service, Supported by LINC+ Project, Sungkyunkwan University

PROFESSIONAL AFFILIATIONS

- 2025 – now The Korean Association of Geographic Information Studies (KAGIS), member
- 2025 – now Society of Photo-Optical Instrumentation Engineers (SPIE), member
- 2024 – now Asia Oceania Geosciences Society (AOGS), member
- 2019 – now European Geosciences Union (EGU), member
- 2019 – now Korea Water Resources Association (KWRA), member
- 2019 – now Korean Meteorological Society (KMS), member
- 2019 – now Korean Society of Civil Engineers (KSCE), member
- 2018 – now American Geophysical Union (AGU), member
- 2018 – now The Korean Society of Remote Sensing (KSRS), member

TECHNICAL SKILLS

LANGUAGES

Proficient in Korean / English
Limited in Japanese

HARDWARE, SOFTWARE, PROGRAMMING LANGUAGE

Matlab, Python, Interactive Data Language (IDL), ENVI, QGIS, R, SNAP
Land Information System Framework (LISF), Community Land Model (CLM)
Sensor techniques (Campbell Scientific CS616/CS625 soil moisture sensors, METER TEROS 12 soil moisture sensor, Hydroinnova CRS-1000 cosmic-ray sensor)

REFERENCES

Dr. Minha Choi (Ph.D. Advisor)

Professor, Civil Engineering & Water Resources, Sungkyunkwan University, Suwon, Republic of Korea
T +82-31-290-7527 | E mhchoi@skku.edu

Dr. Kyung Soo Jun (Ph.D. Committee Chair)

Professor, Civil Engineering & Water Resources, Sungkyunkwan University, Suwon, Republic of Korea
T +82-31-290-7515 | E ksjun@skku.edu; ksjun123@gmail.com

Dr. Daeha Kim

Associate Professor, Department of Civil Engineering, Jeonbuk National University, Jeonju, Republic of Korea
T +82-63-270-2426 | E daeha.kim@jbnu.ac.kr; daeha.kim@gmail.com

Dr. Jongmin Park (Ph.D. Committee)

Assistant Professor, Civil and Environmental Engineering, Korea National University of Transportation, Chungju, Republic of Korea
T +82-43-841-5352 | E jmpark1@ut.ac.kr

Dr. Yonghwan Kwon (Ph.D. Committee)

Data Assimilation Team, Korea Institute of Atmospheric Prediction Systems, Seoul, Republic of Korea
T +82-2-6480-6456 | E yhkwon@kiaps.org